

FGFR2 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes FGFR2 gene fusions from the msk_impact_50k_2026 study, covering 136 fusion events. 108/134 fusions (80.6%) were in-frame. 115/134 fusions (85.8%) retained the Protein kinase domain. Domain retention was tested with Fisher's exact test ($p=0.000424687$, statistically significant at $\alpha=0.05$). No literature benchmark is configured for FGFR2.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 136 analyzed fusion events, identifying 78 distinct fusion partner genes. The most frequent partner was BICC1, observed in 28 events.

Domain retention

Protein kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 99/108 (91.7%) of in-frame fusions retained the domain, $p=0.000424687$ (statistically significant at $\alpha=0.05$). The Protein kinase domain appears to be required for retention. A breakpoint-position permutation test produced a corroborating empirical p-value of 0.000999001.

Domain disruption

Disruption of the Immunoglobulin D1 domain was tested with Fisher's exact test comparing in-frame fusions against all others; 9/108 (8.3%) of in-frame fusions disrupted the domain, $p=0.989695$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.808719.

Cutpoint detection

Cutpoint detection scanned 134 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 767 aa (exon 17; permutation-corrected $p=0.000999001$, statistically significant at $\alpha=0.05$). This is 10 aa from the nearest configured domain boundary at 757 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" ($n=115$) and "not_retained" ($n=19$). For Frame_status, retained: 99/102 (97.1%, 95% CI 91.7%-99.0%); not_retained: 9/14 (64.3%, 95% CI 38.8%-83.7%). Welch's t-test comparing Tumor_variant_count between groups gave means of 104.965 vs 61.8947, $p=0.20677$ (not statistically significant at $\alpha=0.05$).

Composite evidence score

Composite evidence score produced results for this run (78 row(s) in composite_evidence_ranking); see the accompanying table.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: No mRNA expression data was available for FGFR2 in this cohort; expression-association analysis was skipped.

Joint partner

Joint partner reported: FGFR2 has no gene_pair configured; joint-partner analysis was skipped.

Mutation cooccurrence

Mutation cooccurrence reported: FGFR2 has no mutual_exclusivity_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

Window detection

Window detection produced results for this run (32 row(s) in window_scan, 19 row(s) in top_windows); see the accompanying table.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
BICC1	28	0.20588235294117646	0.30004340774793187	0.009220228469391804	0.9968060394889663	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3620511805257468	1
TACC2	8	0.058823529411764705	0.30004340774793187	0.009220228469391804	0.9928861788617886	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3173455543728466	2
KIAA1217	4	0.029411764705882353	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3095890981318136	3
NOL4	4	0.029411764705882353	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3095890981318136	4
SHTN1	3	0.022058823529411766	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.30738321577887245	5
TBC1D4	3	0.022058823529411766	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.30738321577887245	6
CTNNA3	3	0.022058823529411766	0.30004340774793187	0.009220228469391804	0.9994579945799458	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3073019149658643	7
BAIAP2L1	2	0.014705882352941176	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3051773334259313	8
NRBF2	2	0.014705882352941176	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3051773334259313	9

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
OPTN	2	0.014705882352941176	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3051773334259313	10
SORBS1	2	0.014705882352941176	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3051773334259313	11
TTC28	2	0.014705882352941176	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3051773334259313	12
WAC	2	0.014705882352941176	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3051773334259313	13
AHCYL1	2	0.014705882352941176	0.30004340774793187	0.009220228469391804	0.9991869918699187	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3050553822064191	14
INA	2	0.014705882352941176	0.30004340774793187	0.009220228469391804	0.9983739837398374	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3049334309869069	15
ABLIM1	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	16
ADAM9	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	17
AMOT	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	18
ATAD1	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	19

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
ATP6V1D	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	20
BICD1	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	21
BNIP2	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	22
BTBD16	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	23
CAT	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	24
CCDC186	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	25
CCDC6	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	26
CCDC73	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	27
CEACAM7	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	28
DDX21	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	29

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
DIABLO	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	30
GAB2	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	31
HSPA12A	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	32
KIF14	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	33
L3MBTL3	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	34
LAMC1	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	35
MARVELD3	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	36
MBIP	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	37
MYH15	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	38
MYH9	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	39

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
MYLK	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	40
MYO18B	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	41
NRAP	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	42
PAH	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	43
PALLD	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	44
PDZRN4	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	45
PKP4	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	46
POC1B	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	47
PPP1R21	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	48
PRDX3	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	49

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
RABGAP1L	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	50
ROCK1	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	51
SEPTIN10	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	52
SH3GLB1	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	53
SHC2	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	54
SKI	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	55
SRRT	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	56
STAT4	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	57
TFEC	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	58
UBP1	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	59

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
VCL	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	60
ZFAT	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	61
ZMYM4	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	1.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3029714510729901	62
CEP112	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	0.9983739837398374	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3027275486339657	63
KLHL29	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	0.9983739837398374	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3027275486339657	64
NID2	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	0.9983739837398374	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3027275486339657	65
RNF145	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	0.9983739837398374	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3027275486339657	66
RPAP3	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	0.9983739837398374	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3027275486339657	67
VPS26A	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	0.9983739837398374	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3027275486339657	68
YTHDF3	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	0.9983739837398374	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3027275486339657	69

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
FOXP1	2	0.014705882352941176	0.30004340774793187	0.009220228469391804	0.9723577235772358	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.30103099196251665	70
NRG3	2	0.014705882352941176	0.30004340774793187	0.009220228469391804	0.9707317073170731	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.30078708952349226	71
KCNH7	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	0.9495934959349593	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.295410475463234	72
ANKRD11	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	0.943089430894309	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.29443486570713645	73
ATE1	3	0.022058823529411766	0.30004340774793187	0.009220228469391804	0.8384823848238483	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2831555735024497	74
ADAM12	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	0.8520325203252033	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2807763291217706	75
PRKAG2	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	None	0.739106710891876	confidence_certainty, domain_disruption, domain_retention, recurrence	0.17996641302704716	76
ST8SIA6	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	None	0.739106710891876	confidence_certainty, domain_disruption, domain_retention, recurrence	0.17996641302704716	77
GRID1	1	0.007352941176470588	0.30004340774793187	0.009220228469391804	0.0	0.739106710891876	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.1529714510729901	78

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
152	0	115	1	18	0.0	0.1417910447761194	0.8483511974119787
469	0	115	2	17	0.0	0.019189765458422176	1.7169303332757584

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
676	0	115	3	16	0.0	0.0024714091878270984	2.607055343103334
732	0	115	5	14	0.0	3.4829019675778254e-05	4.458058749354329
767	111	4	5	14	77.7	8.867691233331173e-12	11.052189437117653
768	111	4	19	0	0.0	1.0	-0.0
798	112	3	19	0	0.0	1.0	-0.0
801	113	2	19	0	0.0	1.0	-0.0
802	114	1	19	0	0.0	1.0	-0.0

domain_disruption tables

frame_domain_contingency_table

9	6
99	20

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
ig_d1	Immunoglobulin D1 domain	134	119	0	15	0.0

domain_retention tables

frame_domain_contingency_table

99	16
9	10

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	134	115	3	16	0.09006270188105643

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
11	117	136	0.8602941176470589

frequency tables

Partner_gene_counts

Partner_gene	Event_count
ABLM1	1
ADAM12	1
ADAM9	1
AHCYL1	2
AMOT	1
ANKRD11	1
ATAD1	1
ATE1	3
ATP6V1D	1
BAIAP2L1	2
BICC1	28
BICD1	1
BNIP2	1
BTBD16	1
CAT	1
CCDC186	1
CCDC6	1
CCDC73	1
CEACAM7	1
CEP112	1
CTNNA3	3
DDX21	1
DIABLO	1
FOXP1	2
GAB2	1
GRID1	1
HSPA12A	1
INA	2
KCNH7	1
KIAA1217	4
KIF14	1
KLHL29	1

Partner_gene	Event_count
L3MBTL3	1
LAMC1	1
MARVELD3	1
MBIP	1
MYH15	1
MYH9	1
MYLK	1
MYO18B	1
NID2	1
NOL4	4
NRAP	1
NRBF2	2
NRG3	2
OPTN	2
PAH	1
PALLD	1
PDZRN4	1
PKP4	1
POC1B	1
PPP1R21	1
PRDX3	1
PRKAG2	1
RABGAP1L	1
RNF145	1
ROCK1	1
RPAP3	1
SEPTIN10	1
SH3GLB1	1
SHC2	1
SHTN1	3
SKI	1
SORBS1	2
SRRT	1
ST8SIA6	1
STAT4	1

Partner_gene	Event_count
TACC2	8
TBC1D4	3
TFEC	1
TTC28	2
UBP1	1
VCL	1
VPS26A	1
WAC	2
YTHDF3	1
ZFAT	1
ZMYM4	1

window_detection tables

window_scan

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
567	767	200	114	20	111	4	148.0	2.0932447975848392e-14	13.67917997948573	EVT-P-0000386-T02-IM5-170, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-6, EVT-P-0003660-T01-IM5-110, ... (114 total)
568	768	200	128	6	111	4	3.264705882352941	0.20175643503262466	0.6951726045385277	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-6, ... (128 total)
598	798	200	129	5	112	3	4.392156862745098	0.14685792730682784	0.8331026055409751	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-6, ... (129 total)
601	801	200	130	4	113	2	6.647058823529412	0.09627287375948652	1.016496064236981	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-6, ... (130 total)
667	767	100	114	20	111	4	148.0	2.0932447975848392e-14	13.67917997948573	EVT-P-0000386-T02-IM5-170, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-6, EVT-P-0003660-T01-IM5-110, ... (114 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
668	768	100	128	6	111	4	3.264705882352941	0.20175643503262466	0.6951726045385277	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-6, ... (128 total)
676	776	100	128	6	111	4	3.264705882352941	0.20175643503262466	0.6951726045385277	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-6, ... (128 total)
698	798	100	128	6	112	3	7.0	0.03706091185523423	1.4310838994039352	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-6, ... (128 total)
701	801	100	129	5	113	2	10.59375	0.020395293438474477	1.6904700418647354	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-6, ... (129 total)
702	802	100	130	4	114	1	21.375	0.008980082239738162	2.0467196860386054	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-6, ... (130 total)
717	767	50	113	21	111	4	235.875	6.788793280568991e-16	15.168207415434466	EVT-P-0000386-T02-IM5-170, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-6, EVT-P-0003660-T01-IM5-110, ... (113 total)
718	768	50	127	7	111	4	5.203125	0.0589345360272314	1.2296301311807238	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-6, ... (127 total)
732	782	50	127	7	111	4	5.203125	0.0589345360272314	1.2296301311807238	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-6, ... (127 total)
742	767	25	111	23	111	4	None	1.5809864431344557e-19	18.8010718541014	EVT-P-0000386-T02-IM5-170, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-6, EVT-P-0003660-T01-IM5-110, ... (111 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
743	768	25	125	9	111	4	9.910714285714286	0.0029970001963060054	2.5233132286077047	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-6, ... (125 total)
748	798	50	126	8	112	3	13.333333333333334	0.0014660378745564228	2.833854809709636	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-6, ... (126 total)
751	801	50	127	7	113	2	20.178571428571427	0.000604871568572282	3.218336828622687	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-6, ... (127 total)
752	802	50	128	6	114	1	40.714285714285715	0.00019007465001354953	3.721075800511466	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-6, ... (128 total)
767	792	25	125	9	111	4	9.910714285714286	0.0029970001963060054	2.5233132286077047	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-6, ... (125 total)
767	817	50	128	6	114	1	40.714285714285715	0.00019007465001354953	3.721075800511466	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-6, ... (128 total)
767	867	100	129	5	115	0	None	3.482901967577825e-05	4.458058749354329	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-6, ... (129 total)
767	967	200	129	5	115	0	None	3.482901967577825e-05	4.458058749354329	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-6, ... (129 total)
768	793	25	14	120	0	115	0.0	3.402611770788853e-15	14.46818759995576	EVT-P-0000644-T01-IM3-30, EVT-P-0003724-T01-IM5-20, EVT-P-0011516-T01-IM5-23, EVT-P-0013241-T01-IM5-111, EVT-P-0013756-T01-IM5-165, ... (14 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
768	818	50	17	117	3	112	0.009566326530612245	2.052274151058129e-12	11.687764624876504	EVT-P-0000644-T01-IM3-30, EVT-P-0003724-T01-IM5-20, EVT-P-0009701-T01-IM5-105, EVT-P-0011516-T01-IM5-23, EVT-P-0013241-T01-IM5-111, ... (17 total)
768	868	100	18	116	4	111	0.01287001287001287	8.867691233331171e-12	11.052189437117653	EVT-P-0000644-T01-IM3-30, EVT-P-0003724-T01-IM5-20, EVT-P-0009701-T01-IM5-105, EVT-P-0011516-T01-IM5-23, EVT-P-0013241-T01-IM5-111, ... (18 total)
768	968	200	18	116	4	111	0.01287001287001287	8.867691233331171e-12	11.052189437117653	EVT-P-0000644-T01-IM3-30, EVT-P-0003724-T01-IM5-20, EVT-P-0009701-T01-IM5-105, EVT-P-0011516-T01-IM5-23, EVT-P-0013241-T01-IM5-111, ... (18 total)
770	820	50	4	130	4	111	None	1.0	-0.0	EVT-P-0009701-T01-IM5-105, EVT-P-0031242-T01-IM6-142, EVT-P-0058539-T01-IM6-101, EVT-P-0060245-T01-IM7-53
795	820	25	4	130	4	111	None	1.0	-0.0	EVT-P-0009701-T01-IM5-105, EVT-P-0031242-T01-IM6-142, EVT-P-0058539-T01-IM6-101, EVT-P-0060245-T01-IM7-53
798	823	25	4	130	4	111	None	1.0	-0.0	EVT-P-0009701-T01-IM5-105, EVT-P-0031242-T01-IM6-142, EVT-P-0058539-T01-IM6-101, EVT-P-0060245-T01-IM7-53
798	848	50	4	130	4	111	None	1.0	-0.0	EVT-P-0009701-T01-IM5-105, EVT-P-0031242-T01-IM6-142, EVT-P-0058539-T01-IM6-101, EVT-P-0060245-T01-IM7-53
798	898	100	4	130	4	111	None	1.0	-0.0	EVT-P-0009701-T01-IM5-105, EVT-P-0031242-T01-IM6-142, EVT-P-0058539-T01-IM6-101, EVT-P-0060245-T01-IM7-53
798	998	200	4	130	4	111	None	1.0	-0.0	EVT-P-0009701-T01-IM5-105, EVT-P-0031242-T01-IM6-142, EVT-P-0058539-T01-IM6-101, EVT-P-0060245-T01-IM7-53

top_windows

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
742	767	25	111	23	111	4	None	1.5809864431344557e-19	18.8010718541014	EVT-P-0000386-T02-IM5-170, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-6, EVT-P-0003660-T01-IM5-110, ... (111 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
717	767	50	113	21	111	4	235.875	6.788793280568991e-16	15.168207415434466	EVT-P-0000386-T02-IM5-170, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-6, EVT-P-0003660-T01-IM5-110, ... (113 total)
768	793	25	14	120	0	115	0.0	3.402611770788853e-15	14.46818759995576	EVT-P-0000644-T01-IM3-30, EVT-P-0003724-T01-IM5-20, EVT-P-0011516-T01-IM5-23, EVT-P-0013241-T01-IM5-111, EVT-P-0013756-T01-IM5-165, ... (14 total)
667	767	100	114	20	111	4	148.0	2.0932447975848392e-14	13.67917997948573	EVT-P-0000386-T02-IM5-170, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-6, EVT-P-0003660-T01-IM5-110, ... (114 total)
768	818	50	17	117	3	112	0.009566326530612245	2.052274151058129e-12	11.687764624876504	EVT-P-0000644-T01-IM3-30, EVT-P-0003724-T01-IM5-20, EVT-P-0009701-T01-IM5-105, EVT-P-0011516-T01-IM5-23, EVT-P-0013241-T01-IM5-111, ... (17 total)
768	868	100	18	116	4	111	0.01287001287001287	8.867691233331171e-12	11.052189437117653	EVT-P-0000644-T01-IM3-30, EVT-P-0003724-T01-IM5-20, EVT-P-0009701-T01-IM5-105, EVT-P-0011516-T01-IM5-23, EVT-P-0013241-T01-IM5-111, ... (18 total)
767	867	100	129	5	115	0	None	3.482901967577825e-05	4.458058749354329	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-6, ... (129 total)
752	802	50	128	6	114	1	40.714285714285715	0.00019007465001354953	3.721075800511466	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-6, ... (128 total)
751	801	50	127	7	113	2	20.178571428571427	0.000604871568572282	3.218336828622687	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-6, ... (127 total)
748	798	50	126	8	112	3	13.333333333333334	0.0014660378745564228	2.833854809709636	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-6, ... (126 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
743	768	25	125	9	111	4	9.910714285714286	0.0029970001963060054	2.5233132286077047	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-6, ... (125 total)
702	802	100	130	4	114	1	21.375	0.008980082239738162	2.0467196860386054	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-6, ... (130 total)
701	801	100	129	5	113	2	10.59375	0.020395293438474477	1.6904700418647354	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-6, ... (129 total)
698	798	100	128	6	112	3	7.0	0.03706091185523423	1.4310838994039352	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-6, ... (128 total)
718	768	50	127	7	111	4	5.203125	0.0589345360272314	1.2296301311807238	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-6, ... (127 total)
601	801	200	130	4	113	2	6.647058823529412	0.09627287375948652	1.016496064236981	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-6, ... (130 total)
598	798	200	129	5	112	3	4.392156862745098	0.14685792730682784	0.8331026055409751	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-6, ... (129 total)
668	768	100	128	6	111	4	3.264705882352941	0.20175643503262466	0.6951726045385277	EVT-P-0000386-T02-IM5-170, EVT-P-0000644-T01-IM3-30, EVT-P-0002675-T01-IM3-4, EVT-P-0002810-T01-IM3-169, EVT-P-0003177-T01-IM5-6, ... (128 total)
795	820	25	4	130	4	111	None	1.0	-0.0	EVT-P-0009701-T01-IM5-105, EVT-P-0031242-T01-IM6-142, EVT-P-0058539-T01-IM6-101, EVT-P-0060245-T01-IM7-53

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0021826-T01-IM6-1	P-0021826-T01-IM6		Breast Cancer	IDC	ANKRD11-FGFR2	ANKRD11	unknown	five_prime	123243799	16	732	True	731	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.9097472924187726	True	Immunoglobulin D1 domain		Protein kinase domain	FGFR2 (NM_000141) - ANKRD11 (NM_013275) rearrangement: t(10;16)(q26.13;q24.3)(chr10:g.123243799::chr16:g.89538175) Transcript Fusion { FGFR2: ANKRD11 }	NA
EVT-P-0013241-T01-IM5-2	P-0013241-T01-IM5		Hepatobiliary Cancer	IHCH	ATP6V1D-FGFR2	ATP6V1D	in-frame	five_prime	123242048	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - ATP6V1D (NM_015994) rearrangement: t(10;14)(q26.13;q23.2)(chr10:g.123242048::chr14:g.67818116) Protein fusion: in frame (FGFR2-ATP6V1D)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0009512-T01-IM5-3	P-0009512-T01-IM5		Hepatobiliary Cancer	IHCH	BICC1-FGFR2	BICC1	in-frame	five_prime	123240449	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			Note: The FGFR2 (NM_000141) - BICC1 (NM_001080512) reciprocal rearrangement event is an inversion which results in the fusion of FGFR2 exons 1-17 with BICC1 exons 3-21 . The fusion is predicted to be in frame and includes the kinase domain of FGFR2. Protein fusion: in frame (FGFR2-BICC1)	NA
EVT-P-0002675-T01-IM3-4	P-0002675-T01-IM3		Hepatobiliary Cancer	IHCH	BICC1-FGFR2	BICC1	in-frame	five_prime	123241096	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			NA Protein fusion: in frame (FGFR2-BICC1)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0029067-T01-IM6-5	P-0029067-T01-IM6		Hepatobiliary Cancer	IHCH	BICC1-FGFR2	BICC1	in-frame	five_prime	123240197	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 fused with BICC1 exons 3-21) : c. 2302-67: FGFR2_c.238-6757:BICC1inv Protein Fusion: in frame {FGFR2: BICC1}	NA
EVT-P-0003177-T01-IM5-6	P-0003177-T01-IM5		Hepatobiliary Cancer	IHCH	BICC1-FGFR2	BICC1	in-frame	five_prime	123242188	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			NA Protein fusion: in frame (FGFR2-BICC1)	NA
EVT-P-0025807-T01-IM6-7	P-0025807-T01-IM6		Hepatobiliary Cancer	IHCH	BICC1-FGFR2	BICC1	in-frame	five_prime	123242181	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 fused to BICC1 exons 3-21): c. 2301+895:FGFR2_c.238-35107:BICC1inv Protein Fusion: in frame {FGFR2: BICC1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0025250-T01-IM6-8	P-0025250-T01-IM6		Hepatobiliary Cancer	IHCH	BICC1-FGFR2	BICC1	in-frame	three_prime	123240491	18	768	True	768	True	exact match	lost	0.0	False		Protein kinase domain; Immunoglobulin D1 domain		FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 fused to BICC1 exons 16-21): c.2302-956:FGFR2_c.2182-297:BI CC1inv Protein Fusion: in frame {BICC1: FGFR2}	NA
EVT-P-0032033-T01-IM6-9	P-0032033-T01-IM6		Hepatobiliary Cancer	IHCH	BICC1-FGFR2	BICC1	in-frame	five_prime	123241685	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - BICC1 (NM_001080512) Fusion (FGFR2 exon 18 fused with BICC1 exon3) : c.2301+1527:FGFR2_c.237+38877:BI CC1inv Protein Fusion: in frame {FGFR2: BICC1}	NA

event_id	sample_id	patient_id	tumor_type	oncotre_e_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000538-T01-IM5-10	P-000538-T01-IM5		Hepatobiliary Cancer	IHCH	BICC1-FGFR2	BICC1	in-frame	five_prime	123242753	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - BICC1 (NM_001080512) reciprocal fusion (FGFR2 exons 1-17 with BICC1 exons 16-21): c.2301+459:FGFR2_c.2182-276:BICC1inv Protein fusion: in frame (FGFR2-BICC1)	NA
EVT-P-0020186-T01-IM6-11	P-0020186-T01-IM6		Hepatobiliary Cancer	IHCH	BICC1-FGFR2	BICC1	in-frame	five_prime	123241148	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exon17 fused with BICC1 exon3): c.2302-1613:FGFR2_c.237+32254:BICC1inv Protein Fusion: in frame {FGFR2: BICC1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frameshift	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0020025-T01-IM6-12	P-0020025-T01-IM6		Cancer of Unknown Primary	ADNOS	BICC1-FGFR2	BICC1	in-frame	five_prime	123241984	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 with BICC1 exons 3-21) : c. 2301+1228:FGFR2_c.238-4773:BICC1inv Protein Fusion: in frame {FGFR2: BICC1}	NA
EVT-P-0018115-T01-IM6-13	P-0018115-T01-IM6		Hepatobiliary Cancer	IHCH	BICC1-FGFR2	BICC1	in-frame	five_prime	123241970	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 fused in-frame with BICC1 exons 3-21): c. 2301+1242:FGFR2_c.237+34947:BICC1dup Protein Fusion: in frame {FGFR2: BICC1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frameshift	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015942-T01-IM6-14	P-0015942-T01-IM6		Hepatobiliary Cancer	IHCH	BICC1-FGFR2	BICC1	in-frame	five_prime	123241020	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - BICC1 (NM_001080512) Fusion (FGFR2 exons 1-17 with BICC1 exons 9-21) : c. 2302-1485:FGFR2_c.1048-60: BICC1inv Protein Fusion: in frame {FGFR2: BICC1}	NA
EVT-P-0008146-T01-IM5-15	P-0008146-T01-IM5		Hepatobiliary Cancer	IHCH	BICC1-FGFR2	BICC1	in-frame	five_prime	123240154	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 fused with BICC1 exons 16-21) : c.2302-619:FGFR2_c.2182-744: BICC1inv Protein fusion: in frame (FGFR2-BICC1)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0037617-T01-IM6-16	P-0037617-T01-IM6		Hepatobiliary Cancer	IHCH	BICC1-FGFR2	BICC1	in-frame	three_prime	123241302	18	768	True	768	True	exact match	lost	0.0	False		Protein kinase domain; Immunoglobulin D1 domain		BICC1 (NM_001080512) - FGFR2 (NM_0010141) rearrangement: c.238-34060:BICC1_c.2302-1767:FGFR2inv Protein Fusion: in frame {BICC1: FGFR2}	NA
EVT-P-0015432-T01-IM6-17	P-0015432-T01-IM6		Pancreatic Cancer	PAAD	CAT-FGFR2	CAT	in-frame	five_prime	123242534	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - CAT (NM_001752) rearrangement: t(10;11)(q26.13;p13)(chr10:g.123242534::chr11:g.34467980) Protein fusion: in frame (FGFR2-CAT)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0027305-T01-IM6-18	P-0027305-T01-IM6		Esophagogastric Cancer	STAD	CCDC6-FGFR2	CCDC6	in-frame	five_prime	123241542	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - CCDC6 (NM_005436) fusion (FGFR2 exons 1-17 fused to CCDC6 exons 4-9): c.2301+1670:FGFR2_c.582+6426:CCDC6del Protein Fusion: in frame {FGFR2: CCDC6}	NA
EVT-P-0027741-T01-IM6-19	P-0027741-T01-IM6		Breast Cancer	IDC	CEACAM7-FGFR2	CEACAM7	unknown	five_prime	123241990	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) rearrangement: t(10;19)(q26.13;q13.2)(chr10:g.123241990::chr19:g.42175729) Transcript Fusion {FGFR2: CEACAM7}	NA
EVT-P-0003724-T01-IM5-20	P-0003724-T01-IM5		Soft Tissue Sarcoma	SYNS	CEP112-FGFR2	CEP112	in-frame	three_prime	123241923	18	768	True	768	True	exact match	lost	0.0	False	Protein kinase domain; Immunoglobulin D1 domain			t(17;10)(q24.1;q26.13)(chr17:g.64023428::chr10:g.123241923) Protein fusion: in frame (CEP112-FGFR2)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0019723-T01-IM6-22	P-0019723-T01-IM6		Esophagogastric Cancer	SPDAC	CTNNA3-FGFR2	CTNNA3	in-frame	five_prime	123243018	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - CTNNA3 (NM_013266) rearrangement: c. 2301+194:FGFR2_c.1885-25982:CTNNA3 del Protein Fusion: in frame {FGFR2: CTNNA3 }	NA
EVT-P-0011516-T01-IM5-23	P-0011516-T01-IM5		Hepatobiliary Cancer	IHCH	CTNNA3-FGFR2	CTNNA3	in-frame	three_prime	123241940	18	768	True	768	True	exact match	lost	0.0	False	Protein kinase domain; Immunoglobulin D1 domain			CTNNA3 (NM_013266) -FGFR2 (NM_000141) Rearrangement : c. 1048-8479:CTNNA3_c.2301+1272:FGFR2dup Protein fusion: in frame (CTNNA3-FGFR2)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0020410-T01-IM6-24	P-0020410-T01-IM6		Hepatobiliary Cancer	IHCH	DDX21-FGFR2	DDX21	in-frame	five_prime	123243048	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - DDX21 (NM_004728) fusion (FGFR2 exons 1-17 fused to DDX21 exons 2-15): c. 2301+164:FGFR2_c.88-860:DDX21inv Protein Fusion: in frame {FGFR2: DDX21}	NA
EVT-P-0018869-T03-IM6-26	P-0018869-T03-IM6		Breast Cancer	IDC	FGFR2-ABLIM1	ABLIM1	in-frame	five_prime	123242601	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - ABLIM1 (NM_002313) fusion: c. 2301+611:FGFR2_c.895-12244:ABLIM1del Protein Fusion: in frame {FGFR2: ABLIM1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0026275-T01-IM6-27	P-0026275-T01-IM6		Soft Tissue Sarcoma	PLLS	FGFR2-ADAM12	ADAM12	unknown	five_prime	123246899	15	676	False	675	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.7075812274368231	True	Immunoglobulin D1 domain		Protein kinase domain	FGFR2 (NM_000141) - ADAM12 (NM_003474) rearrangement: c. 2026:FGFR2_c.1618+5260:ADAM12dup Protein Fusion: mid-exon {FGFR2:ADAM12}	NA
EVT-P-0020164-T01-IM6-28	P-0020164-T01-IM6		Hepatobiliary Cancer	IHCH	FGFR2-ADAM9	ADAM9	in-frame	five_prime	123242738	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - ADAM9 (NM_003816) fusion: t(8;10(p11.22;q26.13)(chr8:g.38901161::chr10:g.123242738) Protein Fusion: in frame {FGFR2:ADAM9}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0018920-T01-IM6-29	P-0018920-T01-IM6		Cancer of Unknown Primary	ADNOS	FGFR2-AHCYL1	AHCYL1	in-frame	five_prime	123242706	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - AHCYL1 (NM_006621) fusion (FGFR2 exons 1-17 fused in-frame with AHCYL1 exons 2-17): t(1;10)(p13.3;q26.13)(chr1:g.110531185::chr10:g.123242706) Protein Fusion: in frame {FGFR2: AHCYL1}	NA
EVT-P-000644-T01-IM3-30	P-000644-T01-IM3		Hepatobiliary Cancer	IHCH	FGFR2-AHCYL1	AHCYL1	in-frame	three_prime	123239745	18	768	True	768	True	exact match	lost	0.0	False	Protein kinase domain; Immunoglobulin D1 domain			NA Protein fusion: in frame (AHCYL1-FGFR2)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0012666-T01-IM5-31	P-0012666-T01-IM5		Hepatobiliary Cancer	IHCH	FGFR2-AMOT	AMOT	in-frame	five_prime	123241762	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - AMOT (NM_001113490) Rearrangement: t(10;X)(q26.13.1;q23)(chr10:g.123241762::chrX:g.112058354) Protein fusion: in frame (FGFR2-AMOT)	NA
EVT-P-0047504-T01-IM6-32	P-0047504-T01-IM6		Breast Cancer	IDC	FGFR2-ATAD1	ATAD1	in-frame	five_prime	123241680	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - ATAD1 (NM_032810) fusion: c.2301+1532:FGFR2_c.831+3993:ATAD1del Protein Fusion: in frame {FGFR2:ATAD1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0017424-T01-IM6-33	P-0017424-T01-IM6		Endometrial Cancer	USC	FGFR2-ATE1	ATE1	out-of-frame	five_prime	12324043	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - ATE1 (NM_001001976) rearrangement: c. 2302-908:FGFR2_c.1379-520:A TE1dup Protein Fusion: out of frame {FGFR2:A TE1}	NA
EVT-P-0063512-T01-IM7-35	P-0063512-T01-IM7		Prostate Cancer	PRAD	FGFR2-ATE1	ATE1	unknown	five_prime	123263336	10	469	False	469	True	exact match	lost	0.0	False	Immunoglobulin D1 domain	Protein kinase domain		FGFR2 (NM_000141) - ATE1 (NM_001001976) rearrangement: c. 1407:FGFR2_c.943-1893:ATE1dup Protein Fusion: mid-exon {FGFR2:ATE1}	NA
EVT-P-0063625-T01-IM7-36	P-0063625-T01-IM7		Endometrial Cancer	USC	FGFR2-ATE1	ATE1	in-frame	five_prime	123242847	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - ATE1 (NM_001001976) fusion: c. 2301+365:FGFR2_c.975+6106:A TE1dup Protein Fusion: in frame {FGFR2:ATE1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0052794-T01-IM6-38	P-0052794-T01-IM6		Hepatobiliary Cancer	CHOL	FGFR2-BAIAP2L1	BAIAP2L1	in-frame	five_prime	123240813	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - BAIAP2L1 (NM_018842) fusion: t(7;10)(q22.1;q26.13)(chr7:g.98001949::chr10:g.123240813) Protein Fusion: in frame {FGFR2: BAIAP2L1}	NA
EVT-P-0059296-T01-IM7-39	P-0059296-T01-IM7		Cancer of Unknown Primary	ADNOS	FGFR2-BAIAP2L1	BAIAP2L1	in-frame	five_prime	123242027	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - BAIAP2L1 (NM_018842) fusion: t(7;10)(q22.1;q26.13)(chr7:g.98029133::chr10:g.123242027) Protein Fusion: in frame {FGFR2: BAIAP2L1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frameshift	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0004520-T01-IM5-41	P-0004520-T01-IM5		Hepatobiliary Cancer	IHCH	FGFR2-BICC1	BICC1	in-frame	five_prime	123240269	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2(NM_000141) - BICC1(NM_001080512) Fusion (FGFR2 exon 17 fused with exon 10 of BICC1) : c.2302-734:FGFR2_c.1180-924:BI CC1inv Protein fusion: in frame (FGFR2-BICC1)	NA
EVT-P-0013174-T01-IM5-42	P-0013174-T01-IM5		Hepatobiliary Cancer	IHCH	FGFR2-BICC1	BICC1	in-frame	five_prime	123240123	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 with BICC1 exons 16-21): c.2302-588:FGFR2_c.2182-284:BI CC1inv Protein fusion: in frame (FGFR2-BICC1)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0049135-T07-IM7-43	P-0049135-T07-IM7		Hepatobiliary Cancer	CHOL	FGFR2-BICC1	BICC1	in-frame	five_prime	123239791	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c. 2302-256:FGFR2_c.238-36290:BI CC1inv Protein Fusion: in frame {FGFR2: BICC1}	NA
EVT-P-0068640-T01-IM7-44	P-0068640-T01-IM7		Hepatobiliary Cancer	CHOL	FGFR2-BICC1	BICC1	in-frame	five_prime	123240444	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c. 2302-909:FGFR2_c.238-18504:BI CC1inv Protein Fusion: in frame {FGFR2: BICC1}	NA
EVT-P-0049135-T03-IM6-45	P-0049135-T03-IM6		Hepatobiliary Cancer	CHOL	FGFR2-BICC1	BICC1	in-frame	five_prime	123239791	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c. 2302-256:FGFR2_c.238-36290:BI CC1inv Protein Fusion: in frame {FGFR2: BICC1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frameshift	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0049133-T04-IM6-46	P-0049133-T04-IM6		Hepatobiliary Cancer	CHOL	FGFR2-BICC1	BICC1	in-frame	five_prime	123242098	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c. 2301+1114:FGFR2_c.237+29879:BICC1inv Protein Fusion: in frame {FGFR2: BICC1}	NA
EVT-P-006885-T01-IM5-48	P-006885-T01-IM5		Hepatobiliary Cancer	IHCH	FGFR2-BICC1	BICC1	in-frame	five_prime	123240587	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - BICC1 (NM_001080512) rearrangement: c. 2302-1052:FGFR2_c.238-5239:BICC1inv Protein fusion: in frame (FGFR2-BICC1)	NA
EVT-P-0051773-T01-IM6-49	P-0051773-T01-IM6		Hepatobiliary Cancer	IHCH	FGFR2-BICC1	BICC1	in-frame	five_prime	123242828	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c. 2301+384:FGFR2_c.237+13961:BICC1inv Protein Fusion: in frame {FGFR2: BICC1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0049133-T02-IM6-50	P-0049133-T02-IM6		Hepatobiliary Cancer	IHCH	FGFR2-BICC1	BICC1	in-frame	five_prime	123242098	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c. 2301+1114:FGFR2_c.237+29879:BICC1inv Protein Fusion: in frame {FGFR2: BICC1}	NA
EVT-P-0053822-T01-IM6-51	P-0053822-T01-IM6		Hepatobiliary Cancer	IHCH	FGFR2-BICC1	BICC1	in-frame	five_prime	123243033	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c. 2301+179:FGFR2_c.238-1932:BICC1inv Protein Fusion: in frame {FGFR2: BICC1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_effect_on_frame
EVT-P-0021327-T01-IM6-52	P-0021327-T01-IM6		Hepatobiliary Cancer	IHCH	FGFR2-BICC1	BICC1	in-frame	five_prime	123240344	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 fused with BICC1 exons 16-21): c.2302-16T13>C13:FGFR2_c.237+32254T>C13: BICC1inv Protein Fusion: in frame {FGFR2: BICC1}	NA
EVT-P-0060245-T01-IM7-53	P-0060245-T01-IM7		Hepatobiliary Cancer	IHCH	FGFR2-BICC1	BICC1	unknown	five_prime	123239377	18	820	False	820	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c.2460:FGFR2_c.1179+1103>C13: BICC1inv Protein Fusion: mid-exon {FGFR2: BICC1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0068033-T01-IM7-54	P-0068033-T01-IM7		Hepatobiliary Cancer	IHCH	FGFR2-BICC1	BICC1	in-frame	five_prime	123239673	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c.2302-138:FGFR2_c.238-40422:BICC1inv Protein Fusion: in frame {FGFR2: BICC1}	NA
EVT-P-0039087-T01-IM6-55	P-0039087-T01-IM6		Hepatobiliary Cancer	IHCH	FGFR2-BICC1	BICC1	in-frame	five_prime	123242059	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 fused with BICC1 exons 16-21) : c.2301+1153:FGFR2_c.2181+133:BICC1inv Protein Fusion: in frame {FGFR2: BICC1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_effect_on_frame
EVT-P-0054191-T01-IM6-56	P-0054191-T01-IM6		Hepatobiliary Cancer	IHCH	FGFR2-BICD1	BICD1	in-frame	five_prime	123242849	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - BICD1 (NM_001714) fusion: t(10;12)(q26.13;p11.21)(chr10:g.123242849::chr12:g.32465127) Protein Fusion: in frame {FGFR2: BICD1}	NA
EVT-P-0061764-T01-IM7-57	P-0061764-T01-IM7		Cancer of Unknown Primary	ADNOS	FGFR2-BNIP2	BNIP2	in-frame	five_prime	123243170	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - BNIP2 (NM_004330) fusion: t(10;15)(q26.13;q22.2)(chr10:g.123243170::chr15:g.59975666) Protein Fusion: in frame {FGFR2: BNIP2}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0020135-T01-IM6-58	P-0020135-T01-IM6		Colorectal Cancer	COAD	FGFR2-BTBD16	BTBD16	out-of-frame	five_prime	123239702	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - BTBD16 (NM_144587) rearrangement: c. 2302-167:FGFR2_c.386-107: BTBD16inv Protein Fusion: out of frame {FGFR2:BTBD16}	NA
EVT-P-0033652-T01-IM6-59	P-0033652-T01-IM6		Non-Small Cell Lung Cancer	LUAD	FGFR2-CCDC186	CCDC186	in-frame	five_prime	123240830	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - CCDC186 (NM_018017) Rearrangement : c. 2302-1295:FGFR2_c.1222-3356:CCDC186del Protein Fusion: in frame {FGFR2:CCDC186}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intro_nic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0060582-T01-IM7-60	P-0060582-T01-IM7		Cancer of Unknown Primary	ADNOS	FGFR2-CCDC73	CCDC73	in-frame	five_prime	123241248	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - CCDC73 (NM_001008391) fusion: t(10;11)(q26.13;p13)(chr10:g.123241248::chr11:g.32747346) Protein Fusion: in frame {FGFR2:CCDC73}	NA
EVT-P-0037574-T01-IM6-61	P-0037574-T01-IM6		Glioma	GNG	FGFR2-CTNNA3	CTNNA3	in-frame	five_prime	123240911	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - CTNNA3 (NM_013266) fusion: c. 2302-1376:FGFR2_c.1884+30653:CTNNA3del Protein Fusion: in frame {FGFR2:CTNNA3}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0044768-T01-IM6-63	P-0044768-T01-IM6		Breast Cancer	IDC	FGFR2-DIABLO	DIABLO	in-frame	five_prime	123243152	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - DIABLO (NM_019887) fusion: t(10;12)(q26.13;q24.31)(chr10:g.123243152::chr12:g.122708063) Protein Fusion: in frame {FGFR2: DIABLO}	NA
EVT-P-0058539-T01-IM6-101	P-0058539-T01-IM6		Colorectal Cancer	COAD	FGFR2-FOXP1	FOXP1	unknown	five_prime	123239434	18	801	False	801	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - FOXP1 (NM_001244814) fusion: t(3;10)(p13;q26.13)(chr3:g.71113500::chr10:g.123239434) Protein Fusion: mid-exon {FGFR2:FOXP1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0061783-T01-IM7-102	P-0061783-T01-IM7		Pancreatic Cancer	PAAD	FGFR2-FOXP1	FOXP1	in-frame	five_prime	123241483	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - FOXP1 (NM_001244814) fusion: t(3;10)(p13;q26.13)(chr3:g.71192573::chr10:g.123241483) Protein Fusion: in frame {FGFR2: FOXP1}	NA
EVT-P-0055215-T01-IM6-103	P-0055215-T01-IM6		Ovarian Cancer	HGSOC	FGFR2-GRID1	GRID1	out-of-frame	five_prime	123323934	4	152	True	151	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False	Immunoglobulin D1 domain	Protein kinase domain		FGFR2 (NM_000141) - GRID1 (NM_017551) rearrangement: c.454+82:FGFR2_c.726+103661:GRID1 del Protein Fusion: out of frame {FGFR2: GRID1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_effect_on_frame
EVT-P-0015437-T01-IM6-104	P-0015437-T01-IM6		Breast Cancer	IDC	FGFR2-HSPA12A	HSPA12A	in-frame	five_prime	123242591	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - HSPA12A (NM_025015) rearrangement: c. 2301+621:FGFR2_c.663+3346:HSPA12A del Protein fusion: in frame (FGFR2-HSPA12A)	NA
EVT-P-0009701-T01-IM5-105	P-0009701-T01-IM5		Cancer of Unknown Primary	CUP	FGFR2-KCNH7	KCNH7	unknown	five_prime	123239444	18	798	False	797	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - KCNH7 (NM_033272) Rearrangement: t(2,10)(q24.3;q26.13) (chr2:g.163263969::chr10:g.123239444) Protein fusion: mid-exon (FGFR2-KCNH7)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0008658-T01-IM5-107	P-0008658-T01-IM5		Hepatobiliary Cancer	IHCH	FGFR2-KIAA1217	KIAA1217	in-frame	five_prime	123241193	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - KIAA1217 (NM_019590) Rearrangement: c. 2302-1658:FGFR2_c.355-22274: KIAA1217inv Protein fusion: in frame (FGFR2-KIAA1217)	NA
EVT-P-0034784-T01-IM6-108	P-0034784-T01-IM6		Hepatobiliary Cancer	IHCH	FGFR2-KIAA1217	KIAA1217	in-frame	five_prime	123242254	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - KIAA1217 (NM_019590) fusion: c. 2301+958:FGFR2_c.355-49659:KIAA1217inv Protein Fusion: in frame {FGFR2: KIAA1217}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0013405-T01-IM5-109	P-0013405-T01-I-M5		Hepatobiliary Cancer	IHCH	FGFR2-KIAA1217	KIAA1217	in-frame	five_prime	123241766	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - KIAA1217 (NM_019590) Rearrangement: c. 2301+146:FGFR2_c.354+73179:KIAA1217inv Protein fusion: in frame (FGFR2-KIAA1217)	NA
EVT-P-0003660-T01-IM5-110	P-0003660-T01-I-M5		Cancer of Unknown Primary	CUP	FGFR2-KIF14	KIF14	in-frame	five_prime	123240038	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			t(1;10)(q32.1;q26.13)(chr1:g.200555931::chr10:g.123240038) Protein fusion: in frame (FGFR2-KIF14)	NA
EVT-P-0013241-T01-IM5-111	P-0013241-T01-I-M5		Hepatobiliary Cancer	IHCH	FGFR2-KLHL29	KLHL29	unknown	three_prime	123241923	18	768	True	768	True	exact match	lost	0.0	False		Protein kinase domain; Immunoglobulin D1 domain		KLHL29 (NM_052920) - FGFR2 (NM_000141) rearrangement: t(2;10)(p24.1;q26.13)(chr2:g.23647979::chr10:g.123241923) Transcript fusion (KLHL29-FGFR2)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0019439-T02-IM6-112	P-0019439-T02-IM6		Hepatobiliary Cancer	HCCIHC	FGFR2-L3MBTL3	L3MBTL3	in-frame	five_prime	123239680	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - L3MBTL3 (NM_032438) rearrangement: t(6;10)(q23.1;q26.1)(chr6:g.130408449::chr10:g.123239680) Protein Fusion: in frame {FGFR2: L3MBTL3}	NA
EVT-P-0011394-T01-IM5-113	P-0011394-T01-IM5		Hepatobiliary Cancer	IHCH	FGFR2-LAMC1	LAMC1	in-frame	five_prime	123240042	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - LAMC1 (NM_002293) rearrangement: t(1;10)(q25.3;q26.13)(chr1:g.183109471::chr10:g.123240042) Protein fusion: in frame (FGFR2-LAMC1)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0064855-T01-IM7-115	P-0064855-T01-IM7		Non-Small Cell Lung Cancer	LUAD	FGFR2-MBIP	MBIP	in-frame	five_prime	123240070	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - MBIP (NM_016586) fusion: t(10;14)(q26.13;q13.3)(chr10:g.123240070::chr14:g.36787130) Protein Fusion: in frame {FGFR2: MBIP}	NA
EVT-P-0030844-T01-IM6-116	P-0030844-T01-IM6		Colorectal Cancer	COAD	FGFR2-MYH15	MYH15	in-frame	five_prime	123240665	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - MYH15 (NM_014981) rearrangement: t(3;10)(q13.13;q26.13)(chr3:g.108134575::chr10:g.123240665) Protein Fusion: in frame {FGFR2: MYH15}	NA
EVT-P-0027623-T01-IM6-117	P-0027623-T01-IM6		Hepatobiliary Cancer	IHCH	FGFR2-MYLK	MYLK	unknown	five_prime	123240959	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) rearrangement: t(3;10)(q21.1;q26.13)(chr3:g.123525612::chr10:g.123240959) Transcript Fusion {FGFR2: MYLK}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_effect_on_frame
EVT-P-0038235-T01-IM6-118	P-0038235-T01-IM6		Hepatobiliary Cancer	IHCH	FGFR2-NOL4	NOL4	in-frame	five_prime	123240418	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - NOL4 (NM_003787) fusion: t(10;18)(q26.13;q12.1)(chr10:g.123240418::chr18:g.31580817) Protein Fusion: in frame {FGFR2: NOL4}	NA
EVT-P-0012423-T02-IM6-119	P-0012423-T02-IM6		Hepatobiliary Cancer	EHCH	FGFR2-NOL4	NOL4	in-frame	five_prime	123242055	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - NOL4 (NM_003787) fusion (FGFR2 exons 1-17 fused in-frame with NOL4 exons 7-11): t(10;18)(q26.13;q12.1)(chr10:g.12324055::chr18:g.31563418) Protein Fusion: in frame {FGFR2: NOL4}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0006328-T02-IM6-120	P-0006328-T02-IM6		Hepatobiliary Cancer	IHCH	FGFR2-NRBF2	NRBF2	in-frame	five_prime	123241246	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - NRBF2 (NM_030759) fusion: c. 2302-1711:FGFR2_c.157-363NRBF2inv Protein Fusion: in frame {FGFR2: NRBF2}	NA
EVT-P-0058339-T01-IM6-121	P-0058339-T01-IM6		Cancer of Unknown Primary	CUPNOS	FGFR2-PAH	PAH	in-frame	five_prime	123240026	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - PAH (NM_000277) fusion: t(10;12)(q26.13;q23.2)(chr10:g.123240026::chr12:g.103294473) Protein Fusion: in frame {FGFR2: PAH}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015500-T01-IM6-122	P-0015500-T01-IM6		Cancer of Unknown Primary	CUP	FGFR2-PALLD	PALLD	in-frame	five_prime	123239869	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - PALLD (NM_001166108) Rearrangement: t(4;10)(q33;q26.13)(chr4:g.169605121::chr10:g.123239869) Protein fusion: in frame (FGFR2-PALLD)	NA
EVT-P-0021448-T03-IM6-123	P-0021448-T03-IM6		Non-Small Cell Lung Cancer	SARCL	FGFR2-PKP4	PKP4	in-frame	five_prime	123242661	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - PKP4 (NM_003628) rearrangement: t(2;10)(q24.1;q26.13)(chr2:g.159395413::chr10:g.123242661) Protein Fusion: in frame {FGFR2:PKP4}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0059669-T01-IM7-125	P-0059669-T01-I M7		Hepatobiliary Cancer	IHCH	FGFR2-POC1B	POC1B	in-frame	five_prime	123242798	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - POC1B (NM_001199777.2) fusion: t(10;12)(q26.13;q21.33)(chr10:g.123242798::chr12:g.89857934) Protein Fusion: in frame {FGFR2: POC1B}	NA
EVT-P-0008657-T01-IM5-126	P-0008657-T01-I M5		Non-Small Cell Lung Cancer	LUAD	FGFR2-PPP1R21	PPP1R21	in-frame	five_prime	123240684	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR1 (NM_000141) - PPP1R21 (NM_152994) Rearrangement : t(2;10)(2p16.3;10q26.13)(chr2:g.48721864::chr10:g.123240684) Protein fusion: in frame (FGFR2-K LRAQ1)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0057247-T02-IM7-127	P-0057247-T02-IM7		Glioma	GB	FGFR2-PRDX3	PRDX3	in-frame	five_prime	123242894	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - PRDX3 (NM_006793) fusion: c. 2301+318:FGFR2_c.36+287:PRDX3del Protein Fusion: in frame {FGFR2: PRDX3}	NA
EVT-P-0011394-T01-IM5-129	P-0011394-T01-IM5		Hepatobiliary Cancer	IHCH	FGFR2-RABGA P1L	RABGA P1L	in-frame	five_prime	123240065	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - RABGAP1L (NM_014857) rearrangement: t(1;10)(q25.1;q26.13)(chr1:g.174863117::chr10:g.123240065) Protein fusion: in frame (FGFR2-RABGAP1L)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0012639-T01-IM5-130	P-0012639-T01-IM5		Hepatobiliary Cancer	IHCH	FGFR2-SEPTIN10	SEPTIN10	unknown	five_prime	123240798	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - SEPT10 (NM_144710) rearrangement: t(2;10)(q13;q26.13)(chr2:g.110325395::chr10:g.123240798) Protein fusion: mid-exon (FGFR2-SEPT10)	NA
EVT-P-0055202-T01-IM6-131	P-0055202-T01-IM6		Hepatobiliary Cancer	IHCH	FGFR2-SH3GLB1	SH3GLB1	in-frame	five_prime	123242019	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - SH3GLB1 (NM_001206651) rearrangement: t(1;10)(p22.3;q26.13)(chr1:g.87191292::chr10:g.123242019) Protein Fusion: in frame {FGFR2: SH3GLB1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0011516-T01-IM5-132	P-0011516-T01-IM5		Hepatobiliary Cancer	IHCH	FGFR2-SHC2	SHC2	in-frame	five_prime	123243182	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - SH2C (NM_012435) Rearrangement: t(10;19)(q26.13;p13.3)(chr10:g.123243182::chr19:g.447779) Protein fusion: in frame (FGFR2-SHC2)	NA
EVT-P-0032152-T01-IM6-133	P-0032152-T01-IM6		Pancreatic Cancer	PAAD	FGFR2-SHTN1	SHTN1	in-frame	five_prime	123241513	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - KIAA1598 (NM_001127211) fusion: c.2301+1699:FGFR2_c.534+751:KIAA1598 Protein Fusion: in frame {FGFR2: KIAA1598}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0057005-T01-IM6-134	P-0057005-T01-I-M6		Esophageal Cancer	EGC	FGFR2-SHTN1	SHTN1	in-frame	five_prime	123239901	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - KIAA1598 (NM_001127211) fusion: c.2302-366:FGFR2_c.534+660:KIAA1598 del Protein Fusion: in frame {FGFR2: KIAA1598}	NA
EVT-P-0064573-T01-IM7-135	P-0064573-T01-I-M7		Endometrial Cancer	UEC	FGFR2-SKI	SKI	in-frame	five_prime	123242232	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - SKI (NM_003036) fusion: t(1;10)(p36.33;q26.13)(chr1:g.2207811::chr10:g.123242232) Protein Fusion: in frame {FGFR2: SKI}	NA
EVT-P-0054244-T01-IM6-136	P-0054244-T01-I-M6		Cancer of Unknown Primary	CUP	FGFR2-SORBS1	SORBS1	in-frame	five_prime	123243201	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - SORBS1 (NM_001034954) fusion: c.2301+11:FGFR2_c.1738-382:SORBS1 del Protein Fusion: in frame {FGFR2: SORBS1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0057277-T01-IM6-137	P-0057277-T01-IM6		Hepatobiliary Cancer	IHCH	FGFR2-SRRT	SRRT	out-of-frame	five_prime	123242651	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - SRRT (NM_015908) fusion: t(7;10)(q22.1;q26.13)(chr7:g.100475766::chr10:g.123242651) Protein Fusion: out of frame {FGFR2:SRRT}	NA
EVT-P-0055975-T01-IM6-139	P-0055975-T01-IM6		Esophagogastric Cancer	ESCA	FGFR2-STAT4	STAT4	in-frame	five_prime	123240312	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - STAT4 (NM_001243835) rearrangement: t(2;10)(q32.3;q26.13)(chr2:g.191936121::chr10:g.123240312) Protein Fusion: in frame {FGFR2:STAT4}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0045185-T01-IM6-141	P-0045185-T01-IM6		Colorectal Cancer	COAD	FGFR2-TACC2	TACC2	in-frame	five_prime	123240014	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - TACC2 (NM_206862) fusion: c. 2302-479:FGFR2_c.8128-2249:TACC2inv Protein Fusion: in frame {FGFR2: TACC2}	NA
EVT-P-0031242-T01-IM6-142	P-0031242-T01-IM6		Breast Cancer	BRCA	FGFR2-TACC2	TACC2	unknown	five_prime	123239432	18	802	False	801	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - TACC2 (NM_206862) fusion: c. 2405:FGFR2_c.7896+237:TACC2inv Protein Fusion: mid-exon {FGFR2:TACC2}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002364-T03-IM6-143	P-0023364-T03-IM6		Colorectal Cancer	COAD	FGFR2-TACC2	TACC2	unknown	five_prime	123241587	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - TACC2 (NM_206862) rearrangement: c.2301+1625:FGFR2_c.34-5469:TACC2dup Antisense Fusion	NA
EVT-P-0024187-T01-IM6-144	P-0024187-T01-IM6		Hepatobiliary Cancer	IHCH	FGFR2-TACC2	TACC2	unknown	five_prime	123240573	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - TACC2 (NM_206862) Fusion (FGFR2 exon 17 with TACC2 exon12) : c.2302-1038:FGFR2_c.8009:TACC2inv Protein Fusion: mid-exon {FGFR2:TACC2}	NA
EVT-P-0050962-T01-IM6-145	P-0050962-T01-IM6		Cervical Cancer	CEAS	FGFR2-TACC2	TACC2	in-frame	five_prime	123241844	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - TACC2 (NM_206862) fusion: c.2301+1368:FGFR2_c.7608+332:TACC2inv Protein Fusion: in frame {FGFR2:TACC2}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0036358-T01-IM6-146	P-0036358-T01-IM6		Hepatobiliary Cancer	IHCH	FGFR2-TACC2	TACC2	in-frame	five_prime	123239743	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - TACC2 (NM_206862) fusion: c. 2302-208:FGFR2_c.7344+468:TACC2inv Protein Fusion: in frame {FGFR2: TACC2}	NA
EVT-P-0057746-T01-IM6-149	P-0057746-T01-IM6		Hepatobiliary Cancer	CHOL	FGFR2-TACC2	TACC2	in-frame	five_prime	123241998	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - TACC2 (NM_206862) fusion: c. 2301+1214:FGFR2_c.8128-402:TACC2inv Protein Fusion: in frame {FGFR2: TACC2}	NA
EVT-P-0057746-T02-IM6-150	P-0057746-T02-IM6		Hepatobiliary Cancer	IHCH	FGFR2-TACC2	TACC2	in-frame	five_prime	123241998	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - TACC2 (NM_206862) fusion: c. 2301+1214:FGFR2_c.8128-402:TACC2inv Protein Fusion: in frame {FGFR2: TACC2}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0022337-T06-IM6-151	P-0022337-T06-IM6		Breast Cancer	IDC	FGFR2-TBC1D4	TBC1D4	in-frame	five_prime	123241723	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - TBC1D4 (NM_014832) fusion: t(10;13)(q26.13;q22.2)(chr10:g.123241723::chr13:g.75946350) Protein Fusion: in frame {FGFR2: TBC1D4}	NA
EVT-P-0022337-T04-IM6-152	P-0022337-T04-IM6		Breast Cancer	BRCA	FGFR2-TBC1D4	TBC1D4	in-frame	five_prime	123241723	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - TBC1D4 (NM_014832) fusion: t(10;13)(q26.13;q22.2)(chr10:g.123241723::chr13:g.75946350) Protein Fusion: in frame {FGFR2: TBC1D4}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0005291-T01-IM5-153	P-0005291-T01-IM5		Hepatobiliary Cancer	IHCH	FGFR2-TFEC	TFEC	unknown	five_prime	123241729	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_022970) - TFEC (NM_012252) Rearrangement: t(10;7)(q26.13;q31.2)(chr10:g.123241729;chr7:g.115644112) Transcript fusion (FGFR2-TFEC)	NA
EVT-P-0016640-T01-IM6-154	P-0016640-T01-IM6		Hepatobiliary Cancer	CHOL	FGFR2-TTC28	TTC28	in-frame	five_prime	123241944	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000142) - TTC28 (NM_001145418) Rearrangement: t(10;22)(q26.11;q12.1)(chr10:g.123241944::chr22:g.28666001) Protein Fusion: in frame {FGFR2:TTC28}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0051558-T01-IM6-155	P-005158-T01-IM6		Hepatobiliary Cancer	CHOL	FGFR2-TTC28	TTC28	in-frame	five_prime	123243158	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - TTC28 (NM_001145418) fusion: t(10;22)(q26.13;q12.1)(chr10:g.123243158::chr22:g.28605071) Protein Fusion: in frame {FGFR2: TTC28}	NA
EVT-P-0029726-T01-IM6-156	P-0029726-T01-IM6		Breast Cancer	IDC	FGFR2-UBP1	UBP1	in-frame	five_prime	123240062	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - UBP1 (NM_001128161) fusion: t(3;10)(p22.3;q26.13)(chr3:g.33449450::chr10:g.123240062) Protein Fusion: in frame {FGFR2: UBP1}	NA
EVT-P-0004089-T01-IM5-158	P-0004089-T01-IM5		Cancer of Unknown Primary	CUPNOS	FGFR2-VCL	VCL	unknown	five_prime	123242749	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - VCL (NM_014000) Rearrangement : c.2301+463: FGFR2_c.1032:VCLinv Protein fusion: mid-exon (FGFR2-VCL)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0059238-T02-IM7-159	P-0059238-T02-IM7		Hepatobiliary Cancer	CHOL	FGFR2-WAC	WAC	in-frame	five_prime	123242066	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - WAC (NM_016628) fusion: c. 2301+1146:FGFR2_c.1437+679: WACinv Protein Fusion: in frame {FGFR2: WAC}	NA
EVT-P-0058031-T01-IM6-160	P-0058031-T01-IM6		Hepatobiliary Cancer	IHCH	FGFR2-WAC	WAC	in-frame	five_prime	123243180	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - WAC (NM_016628) fusion: c. 2301+32:FGFR2_c.382-351:WACinv Protein Fusion: in frame {FGFR2: WAC}	NA
EVT-P-0025807-T01-IM6-161	P-0025807-T01-IM6		Hepatobiliary Cancer	IHCH	FGFR2-YTHDF3	YTHDF3	unknown	three_prime	123242317	18	768	True	768	True	exact match	lost	0.0	False		Protein kinase domain; Immunoglobulin D1 domain		YTHDF3 (NM_152758) - FGFR2 (NM_000141) rearrangement: t(8;10)(q12.3;q26.13)(chr8:g.64081967::chr10:g.123242317) Protein Fusion: mid-exon {YTHDF3:FGFR2}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0056399-T01-IM6-162	P-0056399-T01-IM6		Esophageal Gastric Cancer	ESCA	FGFR2-ZFAT	ZFAT	in-frame	five_prime	123240282	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - ZFAT (NM_020863) rearrangement: t(8;10)(q24.22;q26.13)(chr8:g.135564146::chr10:g.123240282) Protein Fusion: in frame {FGFR2: ZFAT}	NA
EVT-P-0042241-T01-IM6-163	P-0042241-T01-IM6		Hepatobiliary Cancer	IHCH	FGFR2-ZMYM4	ZMYM4	in-frame	five_prime	123242889	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - ZMYM4 (NM_005095) fusion: t(1;10)(p34.3;q26.13)(chr1:g.35833702::chr10:g.123242889) Protein Fusion: in frame {FGFR2: ZMYM4}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0016195-T01-IM6-164	P-0016195-T01-I-M6		Hepatobiliary Cancer	IHCH	GAB2-FGFR2	GAB2	in-frame	five_prime	123242812	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - GAB2 (NM_080491) fusion (FGFR2 exons 1-17 fused with GAB2 exons 2-10) : t(10;11)(q26.11;q14.1)(chr10:g.123242812::chr11:g.78030553) Protein Fusion: in frame {FGFR2: GAB2}	NA
EVT-P-0013756-T01-IM5-165	P-0013756-T01-I-M5		Cancer of Unknown Primary	CUP	INA-FGFR2	INA	in-frame	three_prime	123242695	18	768	True	768	True	exact match	lost	0.0	False	Protein kinase domain; Immunoglobulin D1 domain			INA (NM_032727) - FGFR2 (NM_000141) Rearrangement : c.1066-3500:INA_c.2301+517:FGFR2inv Protein fusion: in frame (INA-FGFR2)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0013756-T03-IM6-166	P-0013756-T03-IM6		Cancer of Unknown Primary	CUP	INA-FGFR2	INA	in-frame	three_prime	123242695	18	768	True	768	True	exact match	lost	0.0	False		Protein kinase domain; Immunoglobulin D1 domain		INA (NM_032727) - FGFR2 (NM_000141) rearrangement: c.1066-3500:1NA_c.2301+517:FGFR2in v Protein Fusion: in frame {INA:FGFR2}	NA
EVT-P-0002810-T01-IM3-169	P-0002810-T01-IM3		Amplified Cancer	PAMPC A	KIAA1217-FGFR2	KIAA1217	in-frame	five_prime	123240874	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			NA Protein fusion: in frame (FGFR2-KIAA1217)	NA
EVT-P-0000386-T02-IM5-170	P-0000386-T02-IM5		Bladder Cancer	BLCA	MARVELD3-FGFR2	MARVELD3	unknown	five_prime	123243101	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			MARVELD3 (NM_001017967) - FGFR2 (NM_000141) rearrangement: t(16;10)(16q22.2;10q26.13) (chr16:g.71660650::chr10:123243101) Antisense fusion	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_effect_on_frame
EVT-P-0024094-T01-IM6-172	P-0024094-T01-IM6		Hepatobiliary Cancer	IHCH	MYH9-FGFR2	MYH9	in-frame	five_prime	123240418	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - MYH9 (NM_002473) rearrangement: t(10;22)(q26.13;q12.3)(chr10:g.123240418::chr22:g.36682296) Protein Fusion: in frame {FGFR2: MYH9}	NA
EVT-P-0032208-T01-IM6-173	P-0032208-T01-IM6		Hepatobiliary Cancer	IHCH	MYO18B-FGFR2	MYO18B	in-frame	five_prime	123241929	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - MYO18B (NM_032608) rearrangement: t(10;22)(q26.13;q12.2)(chr10:g.123241929::chr22:g.26280218) Protein Fusion: in frame {FGFR2: MYO18B}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0021975-T01-IM6-175	P-0021975-T01-IM6		Breast Cancer	IDC	NID2-FGFR2	NID2	out-of-frame	three_prime	123240637	18	768	True	768	True	exact match	lost	0.0	False		Protein kinase domain; Immunoglobulin D1 domain		NID2 (NM_007361) - FGFR2 (NM_000141) rearrangement: t(10;14)(q26.13;q22.1)(chr10:g.123240637::chr14:g.52482743) Protein Fusion: out of frame {NID2:FGFR2}	NA
EVT-P-0012423-T01-IM5-176	P-0012423-T01-IM5		Hepatobiliary Cancer	CHOL	NOL4-FGFR2	NOL4	in-frame	five_prime	123242054	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - NOL4 (NM_003787) Rearrangement: t(10;18)(q26.13;q12.1)(chr10:g.123242054::chr18:g.31563418) Protein fusion: in frame (FGFR2-NOL4)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0023103-T01-IM6-177	P-0023103-T01-IM6		Hepatobiliary Cancer	IHCH	NOL4-FGFR2	NOL4	in-frame	five_prime	123241921	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - NOL4 (NM_003787) fusion (FGFR2 exons 1-17 fused to NOL4 exons 10-11): t(10;18)(q26.13;q12.1)(chr10:g.123241921::chr18:g.31518701) Protein Fusion: in frame {FGFR2: NOL4}	NA
EVT-P-0008693-T01-IM5-178	P-0008693-T01-IM5		Hepatobiliary Cancer	IHCH	NRAP-FGFR2	NRAP	in-frame	five_prime	123242277	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - NRAP (NM_198060) rearrangement: c.2301+935:FGFR2_c.2458-427:NRAP del Protein fusion: in frame (FGFR2-NRAP)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0006328-T01-IM5-179	P-0006328-T01-IM5		Hepatobiliary Cancer	IHCH	NRBF2-FGFR2	NRBF2	in-frame	five_prime	123241245	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - NRBF2 (NM_030759) rearrangement: c.2302-1710:FGFR2_c.157-364:NRBF2inv Protein fusion: in frame (FGFR2-NRBF2)	NA
EVT-P-0049071-T01-IM6-180	P-0049071-T01-IM6		Non-Small Cell Lung Cancer	LUSC	NRG3-FGFR2	NRG3	out-of-frame	three_prime	123244882	17	732	True	732	True	exact match	disrupted	0.09386281588447654	True		Immunoglobulin D1 domain	Protein kinase domain	NRG3 (NM_001010848) - FGFR2 (NM_000141) rearrangement: c.823+108972:NRG3_c.2195+27:FGFR2inv Protein Fusion: out of frame {NRG3:FGFR2}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0022337-T01-IM6-181	P-0022337-T01-IM6		Breast Cancer	IDC	NRG3-FGFR2	NRG3	out-of-frame	three_prime	123241723	18	768	True	768	True	exact match	lost	0.0	False		Protein kinase domain; Immunoglobulin D1 domain		NRG3 (NM_001010848) - FGFR2 (NM_0010141) Rearrangement: c.953+898-77:NRG3_c.2301+1489:FGFR2inv Protein Fusion: out of frame {NRG3:FGFR2}	NA
EVT-P-0032171-T03-IM6-182	P-0032171-T03-IM6		Ovarian Cancer	LGSOC	OPTN-FGFR2	OPTN	in-frame	five_prime	123239789	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - OPTN (NM_021980) rearrangement: c.2302-254:FGFR2_c.553-1543:OPTNinv Protein Fusion: in frame {FGFR2:OPTN}	NA
EVT-P-0032171-T01-IM6-183	P-0032171-T01-IM6		Ovarian Cancer	LGSOC	OPTN-FGFR2	OPTN	in-frame	five_prime	123239789	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - OPTN (NM_021980) rearrangement: c.2302-254:FGFR2_c.553-1543:OPTNinv Protein Fusion: in frame {FGFR2:OPTN}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frameshift	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0022781-T01-IM6-184	P-0022781-T01-IM6		Ovarian Cancer	HGSOC	PDZRN4-FGFR2	PDZRN4	in-frame	five_prime	123241997	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - PDZRN4 (NM_001164595) rearrangement: t(6;10)(q23.1;q26.1)(chr6:g.130408449::chr10:g.123239680) Protein Fusion: in frame {FGFR2: PDZRN4}	NA
EVT-P-0044768-T01-IM6-187	P-0044768-T01-IM6		Breast Cancer	IDC	RNF145-FGFR2	RNF145	out-of-frame	three_prime	123243127	18	768	True	768	True	exact match	lost	0.0	False	Protein kinase domain; Immunoglobulin D1 domain			RNF145 (NM_144726) - FGFR2 (NM_000141) rearrangement: t(5;10)(q33.3;q26.13)(chr5:g.158598146::chr10:g.123243127) Protein Fusion: out of frame {RNF145:FGFR2}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0006235-T01-IM5-188	P-0006235-T01-IM5		Hepatobiliary Cancer	IHCH	ROCK1-FGFR2	ROCK1	in-frame	five_prime	123240133	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - ROCK1 (NM_005406) rearrangement: t(10;18)(q26.13;q11.1)(chr10:g.123240133::chr18:g18680051) Protein fusion: in frame (FGFR2-ROCK1)	NA
EVT-P-0039370-T01-IM6-189	P-0039370-T01-IM6		Hepatobiliary Cancer	IHCH	RPAP3-FGFR2	RPAP3	in-frame	three_prime	123242832	18	768	True	768	True	exact match	lost	0.0	False	Protein kinase domain; Immunoglobulin D1 domain			RPAP3 (NM_024604) - FGFR2 (NM_000141) rearrangement: t(10;12)(q26.13;q13.11)(chr10:g.123242832::chr12:g.48092574) Protein Fusion: in frame {RPAP3:FGFR2}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frameshift	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0025074-T01-IM6-190	P-0025074-T01-IM6		Cancer of Unknown Primary	ADNOS	SHTN1-FGFR2	SHTN1	in-frame	five_prime	123240122	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - KIAA1598 (NM_001127211) Fusion (FGFR2 exon 17 with KIAA1598 exon 5): c.2302-587:FGFR2_c.267+2596:KIAA1598 Protein Fusion: in frame {FGFR2: KIAA1598}	NA
EVT-P-0032507-T01-IM6-191	P-0032507-T01-IM6		Hepatobiliary Cancer	IHCH	SORBS1-FGFR2	SORBS1	unknown	five_prime	123242268	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) Rearrangement : c.2301+944:FGFR2_chr10:g.97250518del Transcript Fusion { FGFR2: SORBS1 }	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0022337-T01-IM6-192	P-0022337-T01-IM6		Breast Cancer	IDC	TBC1D4-FGFR2	TBC1D4	in-frame	five_prime	123241723	17	767	True	767	True	exact match	retained	1.0	False	Protein kinase domain; Immunoglobulin D1 domain			FGFR2 (NM_000141) - TBC1D4 (NM_014832) Rearrangement : t(10;13)(q26.13;q22.2)(chr10:g.123241723::chr13:g.75946348)	NA
EVT-P-0021788-T01-IM6-193	P-0021788-T01-IM6		Breast Cancer	IDC	VPS26A-FGFR2	VPS26A	in-frame	three_prime	123242226	18	768	True	768	True	exact match	lost	0.0	False	Protein kinase domain; Immunoglobulin D1 domain			VPS26A (NM_004896) - FGFR2 (NM_000141) Rearrangement : c.4-1035:VPS26A_c.2301+986:FGFR2inv Protein Fusion: in frame {VPS26A:FGFR2}	NA

domain retention outliers

FGFR2 domain-retention track



fusion schematic

FGFR2 fusion-transcript schematic

Fusion order varies by row; follow each row's 5' and 3' labels (FGFR2 portion is domain-colored, exon ticks). Partner-gene blocks share one color; row labels identify each partner.



Partner-block colors are decorative; row labels identify partners.

Showing the top 28 of 84 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

reference comparison

Reference vs msk_impact_50k_2026

